

Hardware Architecture — Advanced RC Car V2

As-Built revision · single 3.3 V logic rail + 12 V motor rail · 4WD skid-steer

What changed from the original design document

- **Power rails simplified to 3.3 V and 12 V only.** The 5 V bus is removed; the LM2596 buck is set to **3.3 V** and feeds all logic, sensors and encoders. 12 V goes directly to the motor drivers.
- **AHT20 temperature/humidity sensor removed** — it was a parts-list mistake and is not used.
- **MPU6050 removed** — the IMU is the **BNO055** only.
- **Motor topology corrected to 4WD skid-steer:** two BTS7960 drivers, each driving **one side's motor pair** (not four independent motors).
- **Encoders powered from 3.3 V** so their pulses feed the ESP32 directly — no level shifting.

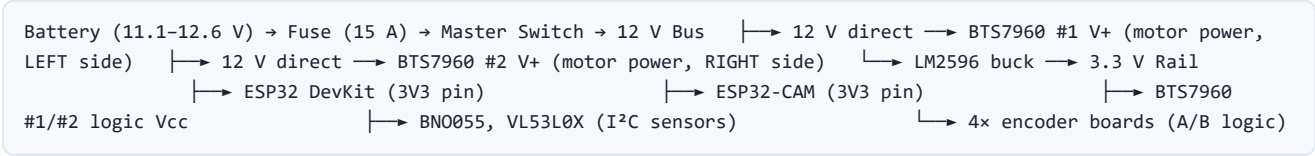
Section 1 — Components (as used)

Component	Qty	Supply	Interface / pins	Role
ESP32 DevKit V1 (ESP-WROOM-32)	1	3.3 V	Wi-Fi AP, PWM, I ² C, UART2, GPIO interrupts	Master MCU — runs the test-bench firmware
ESP32-CAM (AI-Thinker)	1	3.3 V	UART2 link to DevKit	Camera streamer (planned subsystem)
OV2640 camera	1	3.3 V (on-CAM)	DVP (internal to ESP32-CAM)	2 MP image sensor
BTS7960 / IBT-2 H-bridge	2	12 V power · 3.3 V logic	RPWM, LPWM, R_EN, L_EN	One per side — drives a motor pair bidirectionally
JGB37-520 geared motor + quadrature encoder	4	12 V motor · 3.3 V encoder	2 motor leads + encoder A/B	Drive wheels with speed/position feedback
BNO055 9-axis IMU	1	3.3 V	I ² C @ 0x28	Fused absolute orientation (heading/roll/pitch)
VL53L0X ToF range sensor	1	3.3 V	I ² C @ 0x29	Distance sensing (planned subsystem)
LM2596 buck converter	1	12 V in → 3.3 V out	—	Single regulated logic rail (set pot to 3.3 V)
3S Li-ion pack (3×18650)	1	11.1–12.6 V	—	Main supply
Fuse + master switch	1	12 V	—	Protection / power cutoff

~~AHT20 temperature/humidity sensor~~ and ~~MPU6050 IMU~~ are intentionally not part of this build.

Section 2 — Power Architecture (two rails)

The 3S pack feeds a fused, switched 12 V bus that splits into exactly two rails:



- **12 V rail:** direct battery to both BTS7960 V+ inputs for maximum motor torque (drivers accept 6–27 V).

- **3.3 V rail:** the LM2596 is adjusted to 3.3 V and powers *everything logic*. The ESP32 boards are fed on their **3V3 pins**, bypassing their on-board 5 V→3.3 V regulators. The BTS7960 logic inputs are 3.3–5 V compatible, so 3.3 V drives them fine.
- **No 5 V bus.** Nothing in this build requires 5 V.

Buck sizing: set the LM2596 to 3.3 V *before* connecting any load and verify under load. The 3.3 V rail draws ≈ 0.7 A peak (see Section 7); the LM2596's 3 A rating leaves ample margin. Add ≥ 470 μ F input and ≥ 220 μ F output bulk capacitors plus 0.1 μ F ceramics for noise.

Feeding ESP32 via 3V3: when powering on the 3V3 pin you bypass the on-board regulator and reverse-protection, so the 3.3 V rail must be clean and never exceed 3.6 V. Do not also power via USB and VIN at the same time.

Grounding & noise

- Single star-ground near the battery negative. Heavy 12 V motor returns use 14–16 AWG; logic/sensor grounds use 22–24 AWG back to the same node.
- 0.1 μ F ceramics at each IC power pin; 100–470 μ F electrolytic on the 12 V bus near the drivers and on the 3.3 V rail near the ESP32.
- Twist motor + encoder wires; keep I²C (4.7 k Ω pull-ups to 3.3 V) away from motor power.

Section 3 — GPIO Planning (ESP32 DevKit master)

GPIO	Function	Dir	Notes
4	BTS7960 #1 RPWM (LEFT forward)	PWM out	20 kHz
5	BTS7960 #1 LPWM (LEFT reverse)	PWM out	20 kHz
18	BTS7960 #1 R_EN (LEFT)	out	HIGH = enable
19	BTS7960 #1 L_EN (LEFT)	out	HIGH = enable
23	BTS7960 #2 RPWM (RIGHT forward)	PWM out	20 kHz
26	BTS7960 #2 LPWM (RIGHT reverse)	PWM out	20 kHz
27	BTS7960 #2 R_EN (RIGHT)	out	HIGH = enable
13	BTS7960 #2 L_EN (RIGHT)	out	HIGH = enable
34 / 35	Encoder 1 A/B — Left-Front	in (IRQ)	input-only, external pull-up to 3.3 V
32 / 33	Encoder 2 A/B — Left-Rear	in (IRQ)	—
36 / 39	Encoder 3 A/B — Right-Front	in (IRQ)	input-only (SVP/SVN), external pull-up
25 / 14	Encoder 4 A/B — Right-Rear	in (IRQ)	—
21 / 22	I ² C SDA / SCL	I/O	BNO055 (0x28), VL53L0X (0x29); 4.7 kΩ pull-ups to 3.3 V
16 / 17	UART2 RX2 / TX2	I/O	link to ESP32-CAM TX0/RX0 (planned)
0 / 2 / 15	Boot strapping	—	left unused for I/O
6–11	SPI flash	—	do not use

3.3 V everywhere means encoder A/B pulses are already at logic level — they connect straight to the ESP32 inputs with no dividers or level shifters.

Section 4 — Drive Topology (4WD skid-steer)

Four motors, two driver boards, two control channels. Each BTS7960 is a single bidirectional H-bridge (RPWM + LPWM + R_EN + L_EN) wired to the **two motors on one side in parallel**. Steering is differential (drive sides at different speeds), like a tank.

Channel	Driver	Motors	Encoders (feedback)
LEFT	BTS7960 #1 (GPIO 4/5/18/19)	Left-Front + Left-Rear (parallel)	Enc 1 (34/35), Enc 2 (32/33)
RIGHT	BTS7960 #2 (GPIO 23/26/27/13)	Right-Front + Right-Rear (parallel)	Enc 3 (36/39), Enc 4 (25/14)

Both motors on a side share one H-bridge, so they always receive the same drive — but each wheel still has its **own** encoder, giving 4 independent feedback channels for diagnosing a slipping or stalled wheel.

Section 5 — Wiring (per component)

BTS7960 #1 — LEFT side

- RPWM→GPIO4, LPWM→GPIO5, R_EN→GPIO18, L_EN→GPIO19
- Vcc (logic)→3.3 V rail · GND→common ground · B+ →12 V bus

- Motor outputs → both LEFT motors wired in parallel (observe polarity so both spin the same way)

BTS7960 #2 — RIGHT side

- RPWM→GPIO23, LPWM→GPIO26, R_EN→GPIO27, L_EN→GPIO13
- Vcc (logic)→3.3 V rail · GND→common ground · B+ →12 V bus
- Motor outputs → both RIGHT motors in parallel

Motors & encoders (×4)

- Motor leads → side's BTS7960 output (12 V pulsed)
- Encoder Vcc→**3.3 V rail** · Encoder GND→ground · A/B→assigned GPIO (Section 3)

Sensors (I²C)

- BNO055: Vcc→3.3 V, GND, SDA→21, SCL→22, ADR/SDO→GND (0x28)
- VL53L0X: Vcc→3.3 V, GND, SDA→21, SCL→22 (0x29), XSHUT→3.3 V to enable

ESP32-CAM (planned)

- 3V3→3.3 V rail · GND→ground · CAM TX0(GPIO1)→DevKit RX2(GPIO16) · CAM RX0(GPIO3)→DevKit TX2(GPIO17)

Section 6 — Signal Flow

ESP32 DevKit → (PWM + R_EN/L_EN) → BTS7960 x2 → (12 V PWM) → motor pairs Wheel encoders → (A/B quadrature, IRQ) → ESP32 DevKit BNO055, VL53L0X → (I²C @ 3.3 V) → ESP32 DevKit ESP32-CAM (OV2640) → (UART2) → ESP32 DevKit [planned] ESP32 DevKit → (Wi-Fi AP, HTTP + WebSocket) → phone test UI

Section 7 — Power Budget

Load	Rail	Typ	Peak	Notes
ESP32 DevKit	3.3 V	80 mA	240 mA	Wi-Fi AP active
ESP32-CAM	3.3 V	180 mA	310 mA	camera + Wi-Fi (planned)
BNO055	3.3 V	12 mA	15 mA	fusion mode
VL53L0X	3.3 V	19 mA	40 mA	active ranging
4× encoders	3.3 V	20 mA	40 mA	hall sensors
2× BTS7960 logic	3.3 V	40 mA	40 mA	gate-drive/logic
3.3 V rail total	3.3 V	≈0.35 A	≈0.7 A	well within LM2596 3 A
4× DC motors	12 V	~2–10 A	~12.8 A	stall ~3.2 A each; motors dominate

- **12 V rail** sized for motor stall: use a 15 A fuse and a pack rated ≥20–30 A burst.
- **3.3 V rail:** LM2596 input current at 12 V for 3.3 V @0.7 A out ≈ 0.25 A; the buck dissipates only ~1 W.
- **Battery:** a 5–6 Ah 3S pack gives ~1.5–2 h moderate runtime.

Section 8 — Firmware Mapping

This hardware is exercised by the `CarTestBench` ESP32 firmware. Each subsystem has its own page on the phone UI (the ESP32 hosts a Wi-Fi access point — connect and browse to its IP):

Page	Hardware tested	Status
<code>/motors</code>	LEFT / RIGHT BTS7960 channels	implemented
<code>/encoders</code>	4 wheel encoders + CPR calibration	implemented
<code>/imu</code>	BNO055 orientation	groundwork (feature-flagged)
Range	VL53L0X	planned
Camera	ESP32-CAM / OV2640	planned

Pin assignments above are the single source of truth and match `config.h` in the firmware (`PINS_LEFT` , `PINS_RIGHT` , `PINS_ENCODER[ ]`).